Fiscal Year:	FY 2017	**Task Last Updated:**	FY 01/24/2018
PI Name:	Buckey, Jay C. M.D.		
Project Title:	Role of the Cranial Venous Circulation in Microgravity-Associated Visual Changes		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Address updated 9/2008		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	08/01/2013	**End Date:**	05/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2017 per NSBRI (Ed., 3/1/16) NOTE: Title change to "Role of the Cranial Venous Circulation in Microgravity-Associated Visual Changes" (original proposal title was "Ocular Venous Contributions to Spaceflight Visual Impairment")--Ed., 2/6/14		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Weaver, John Ph.D. (Dartmouth College) Knaus, Darin Ph.D. (Creare, Inc.) Deserranno, Dimitri Ph.D. (Creare, Inc.) Belden, Clifford M.D. (Dartmouth College) Kattamis, Nicholas Ph.D. (Creare, Inc.) Phillips, Scott Ph.D. (Creare, Inc.) Davis, Brynmor Ph.D. (Creare, Inc.) Zegans, Michael M.D. (Dartmouth College)		
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Performance Goal Text:			

Task Description:	We developed a numerical model of the cerebral venous circulation that describes how this circulation responds to both fluid shifts and changes in gravitational forces. We validated this model using physiologic studies to measure the responses of the cranial vascular and cerebrospinal fluid (CSF) systems to fluid shifts in both the supine and prone positions. This model and supporting data will provide a means to develop hypotheses about how microgravity produces visual changes over time and may allow predictions about which subjects may be at risk for the visual deficits associated with microgravity. Aim #1: Develop a numerical model to estimate changes in intracranial venous flow, volume, compliance, and pressure in response to a fluid shift and changes in hydrostatic gradients. Include tissue compressive forces in the model. We developed a numerical model that predicts alterations in the hemodynamics of the cranial venous system produced by changes in hydrostatic gradients and fluid shifts. While models describing the cranial venous system exist in the literature, incorporation of hydrostatic gradient and fluid shift effects is novel. Our computer model consists of two parts—a system model and a structural model. The system model describes flow and pressure in the circulatory system, the cerebral spinal fluid regulatory system, and the eye's aqueous humor regulatory system. The structural model uses a separate decoupled finite element model to describe the eye globe and structures contained within. The structural model was developed on a separate parallel effort funded by the NASA EPSCoR (Established Program to Stimulate Competitive Research) program. Both the system level model and the structural model account for changes in tissue properties that may be induced in microgravity, as well as effects of microgravity exposure duration. We validated the computer model with magnetic resonance imaging (MRI) measurements of venous flow, venous volume, venous pressure, intracranial compliance, CSF volume, and flow pulsatility during cephalad fluid shifts both supine and prone. We also used measurements of intraocular pressure (IOP) and other ocular parameters obtained outside of the MRI magnet. This model and supporting data will provide a means to develop hypotheses about how microgravity produces visual changes over time and may allow predictions about which subjects may be at risk for visual deficits associated with microgravity. Aim #2: Determine the cranial venous changes produced by fluid shifts and altered hydrostatic gradients. Use interventions that can produce fluid shifts (lower body negative pressure and lower body positive pressure) and alter hydrostatic gradients (supine and prone postures). These experiments are designed to provide data for validating and verifying the model developed as a part of Aim #1. During this reporting period, we published the results on the effects of posture on eye parameters in the prone and supine postures over 60 minutes. These data are essential for understanding the effects of changes in the gravitational vector on the eye. We completed studies on 14 individuals both inside and outside of the MR device in both the supine and prone postures. MR Imaging was also done with and without lower body negative pressure (LBNP) and lower body positive pressure (LBPP). These data have been incorporated into the model and are being prepared for publication. Aim #3: Identify individuals with common intracranial venous variants, and study them using the protocol outlined in Aim #2. Our cohort of subjects included subjects with a range of venous anatomy, which was adequate for our purposes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The model and experimental results will give a better understanding of cranial venous insufficiency. Cranial venous insufficiency results when venous outflow from the head is reduced or obstructed. This can be due to an increase in venous resistance from anatomical variations. The increased resistance to flow can produce headaches or vision changes. Venous insufficiency has been proposed as a possible etiology for symptoms in acute mountain sickness, obstructive sleep apnea, jugular outflow obstruction syndrome, multiple sclerosis, and idiopathic intracranial hypertension (IIH). Bilateral transverse sinus stenosis is found in 90% of IIH sufferers, and internal jugular vein stenosis occurs in 80% of IIH patients. Although numerical models of the circulatory system, cerebral venous system, and cerebral spinal fluid system exist (as do fluid models of the aqueous humor regulatory system and structural finite element models of the eye) the model developed on this project is the first comprehensive model that links the effects of all systems together. It could be used in the future to predict how surgical interventions, such as sagittal stenting, can improve an individual's symptoms. The model is also useful for understanding the visual problems that occur during prolonged surgical procedures in unusual postures (i.e., robotic prostatectomy, spinal surgery).
Task Progress:	Aim #1: Develop a numerical model to estimate changes in intracranial venous flow, volume, compliance, and pressure in response to a fluid shift and changes in hydrostatic gradients. Include tissue compressive forces in the model. 1. Built a comprehensive model of the cerebrovascular circulation that includes a circulatory sub-model, an aqueous humor submodel, and a cerebrospinal fluid submodel. 2. Model includes effects of hydrostatic gradients and tissue weight, making it well suited for predicting microgravity effects. 3. Model validated using physiologic data from the lab studies that examined the effects of posture (prone vs. supine) and fluid shifts (LBNP) on the cerebrovascular circulation using detailed measurements of the eye and cerebrovascular circulation using various ocular measures (optical coherence tomography, optical biometry) and MRI measures of arterial, venous, and CSF flow. 4. Overall model description and outputs for sample microgravity conditions being prepared for publication. Aim #2: Determine the cranial venous changes produced by fluid shifts and altered hydrostatic gradients. Use interventions that can produce fluid shifts (lower body negative pressure and lower body positive pressure) and alter hydrostatic gradients (supine and prone postures). These experiments are designed to provide data for validating and verifying the model developed as a part of Aim #1. 1. Completed a study on the short-term effects of posture and microgravity (using parabolic flight) on the eye and published the results. 2. Completed a study on the longer-term (60 minutes) effects of posture (changes in direction of hydrostatic gradients going from supine to prone) and fluid shifts on the eye and published the results. 3. Completed a study on the effects of posture/changes in the direction of hydrostatic gradients and fluid shifts on the cerebrovascular circulation and CSF using MR imaging. Results are being prepared for publication.

	Aim #3: Identify individuals with common intracranial venous variants, and study them using the protocol outlined in Aim #2. 1. Subjects studied for this project had a range of venous anatomy that is suitable for further analysis.
Bibliography Type:	Description: (Last Updated: 05/15/2026)
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